

Shells and cases

A shell functions as both a skeleton and a mobile home. It keeps an animal in shape and helps protect the soft parts of its body. Shells also stop animals from drying out on land, including sea creatures that are exposed at low tide. The most common shelled animals are mollusks. They take chalky minerals out of the water around them, or from their food, then deposit them in layers, building up a thick, strong shell.



Twisted homes

Snails and their relatives usually have coiled shells with a clockwise twist. Extra turns are added to the shell as the snail grows, so that the animal always has enough space to pull its body safely inside.

Hinged shields

Scallops and other bivalves are mollusks with a two-part shell. These two parts, or valves, are held together by strong muscles and a flexible hinge. When bivalves feed, they open up their shells, but if danger threatens, most kinds close up tight. Scallops react differently. By clapping the two halves of their shells together and forcing out jets of water, they can swim to safety in rapid jerks.

Layer of tissue, or mantle, lining shell

Hidden chambers

A nautilus has a spiral-shaped shell, but with a different internal plan from a snail's. It is divided into separate compartments sealed by partition walls. The animal lives in the largest of these, which opens to the outside. The other shell compartments are filled with gas, and the nautilus uses them to adjust its depth in the water when swimming.



Chain mail

Chitons are unusual mollusks with flat shells made up of eight overlapping plates called valves. Surrounded by a rubbery "skirt," the valves are connected by hinges, which allow a chiton to bend as it slides over rocks. If a chiton is dislodged, it can curl up in its chain-mail shell to protect its soft underside.

